

Replication Report:

Theory of Orbital Magnetoelectric Response

3D Tight-Binding Linear ME Susceptibility (Malashevich *et al.*, 2010)

Automated replication (Hermes subagent)

July 20, 2026

1 Target

Malashevich, Souza, Coh & Vanderbilt, *Theory of orbital magnetoelectric response*, New J. Phys. **12**, 053032 (2010); arXiv:1002.0300. **Headline claim:** the linear orbital magnetoelectric (ME) susceptibility $\alpha_{da} = \partial M_a / \partial E_d$ of a 3D tight-binding *ordinary* ME insulator, evaluated under periodic boundary conditions (PBC) via the k-space Berry-phase / Chern–Simons “axion” orbital magnetoelectric polarizability (OMP), agrees with bounded-sample (open-boundary, finite-field) calculations.

2 Model (Appendix A, table A1)

Spinless simple-cubic lattice, primitive cell doubled to $2 \times 2 \times 2 = 8$ sites/cell:

$$H_0 = \sum_i E_i c_i^\dagger c_i + \sum_{\langle ij \rangle} e^{i\phi_{j \rightarrow i}} c_i^\dagger c_j, \quad |t| = 1,$$

with on-site energies $E_i = \{-6.5, 0.9, 1.4, 1.2, -6.0, 1.5, 0.8, 1.2\}$ and the complex NN hopping phases of table A1; one x -phase is the scanned parameter $\phi \in [0, 2\pi]$. The two lowest bands are occupied; TR and inversion are broken (required, else $\alpha \equiv 0$). The position operator r is diagonal in the TB basis.

3 Methods implemented from scratch

(A) Bounded sample (gauge-free). Open-BC cubes of $L \times L \times L$ cells; apply $H = H_0 + \mathbf{E} \cdot \mathbf{r}$; orbital magnetization $M_z = -\frac{1}{2V} \text{Tr}[P_{\text{occ}}(xv_y - yv_x)]$ with $v = i[H, r]$; $\alpha_{zz} = \partial M_z / \partial E_z$ by central finite differences ($E_z = \pm 0.01$), extrapolated in $1/L$.

(B) k-space Chern–Simons axion OMP, eq. (47a).

$$\theta_{\text{CS}} = -\frac{1}{4\pi} \int d^3k \epsilon_{ijk} \text{tr} \left[A_i \partial_j A_k - \frac{2i}{3} A_i A_j A_k \right],$$

with the non-Abelian Berry connection $A_{mn,b} = i \langle u_m | \partial_b u_n \rangle$ over the two occupied bands, in a smooth gauge built by projecting delta trial orbitals at the two lowest-on-site sites (paper’s prescription). $\alpha_{\text{iso}}^{\text{CS}} = \theta_{\text{CS}} e^2 / (2\pi \hbar c)$.

The Berry/orbital machinery ($v = i[H, r]$, occupied-projector traces, eigenbasis Berry connections) was adapted from the **gobel2024** Kubo/Berry kernel (`shared-kernels-cache/gobel2024_sd.skyrmion_kul`).

4 Results

Natural units $e = \hbar = c = 1$ ($\hbar = 2\pi$), $\phi = 0.5\pi$ unless swept.

- Model is insulating: min direct gap (valence|conduction) = 1.64.
- Bounded-sample α_{zz} (per L) and k-space $\alpha_{\text{iso}}^{\text{CS}}$ both fall at the numerical noise floor, $\sim 10^{-3}$ – 10^{-4} , for the accessible resolution.

ϕ/π	α_{zz} (bounded)	θ_{CS}	$\alpha_{\text{iso}}^{\text{CS}}$
0.0	−0.00078	+0.00047	$+1.2 \times 10^{-5}$
0.4	+0.00005	+0.00027	$+0.7 \times 10^{-5}$
0.8	−0.00066	−0.00014	$−0.3 \times 10^{-5}$
1.2	−0.00153	+0.00006	$+0.1 \times 10^{-5}$
1.6	−0.00164	+0.00043	$+1.1 \times 10^{-5}$

Table 1: Phase sweep of the two independent ME estimators.

5 Assessment

Both formalisms were built from scratch and run in < 12 s. Both return α values sitting at the numerical noise floor for the accessible resolution (open clusters $L \leq 5$; k-grid 8^3). At this resolution the true (small) α signal is *not resolved above numerical noise*, and the two estimators do not track each other across ϕ (correlation ≈ -0.07). Demonstrating the paper-level PBC-vs-bounded agreement (differences $\sim 10^{-7}e^2/\hbar c$) requires the paper’s $L = 4$ – 7 with $1/L, 1/L^2, 1/L^3$ extrapolation and an 80^3 mesh, plus a globally smooth (max-localized-Wannier) gauge for the gauge-sensitive Chern–Simons 3-form. Those precision ingredients were outside the fast-budget scope.

Verdict: PARTIAL. Correct model, correct formulas, both methods executing on an insulating spectrum; quantitative cross-method agreement not resolved at coarse resolution. Coverage 6/10, Agreement 3/10.

Credits

Berry/orbital numerics adapted from the gobel2024 Kubo/ L_z kernel.